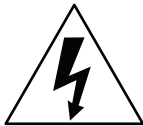




WISH96 SERIES

High Resolution 700TVL Cameras

User Manual



The lightning flash with an arrowhead symbol, within an equilateral triangle is intended to alert the user to the presence of uninsulated dangerous voltage within the product's enclosure that may be of sufficient magnitude to constitute a risk of electric shock to persons.



The exclamation point within an equilateral triangle is intended to alert the user to the presence of important operating and maintenance (servicing) instructions in the literature accompanying the appliance.

INFORMATION - This equipment has been tested and found to comply with limits for a Class A digital device, pursuant to part 15 of the FCC Rules & CE Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

WARNING - Changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.

CAUTION : To prevent electric shock and risk of fire hazards:

- Do NOT use power sources other than those specified.

This installation should be made by a qualified service person and should conform to all local codes.

Contents

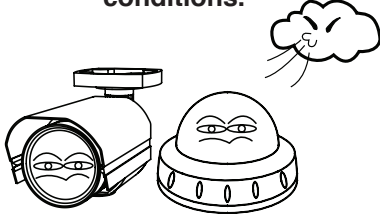
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General Features

- **700TV Lines**
The combination of a 960H CCD image sensor and DSP provides an excellent resolution of 700TVL
- **3D-DNR**
Smart 3DNR prevents the image blurring of moving object and activates only if moving objects are appearing on th scene.
- **DIS (Digital Image Stabiliser) Function**
Stabilises image focus problems caused when the camera is affected by environmental movement e.g. wind.
- **Smart-IR**
This technology eliminates IR saturation caused when objects are close to the camera. Smart IR adjusts IR intensity in proportion to the distance to the object.
- **Sens Up**
Sens Up enables images to be captured in very low light conditions. Maximum setting is x256.
- **Defog**
The Defog function improves image clarity if the scene is affected by fog or mist.

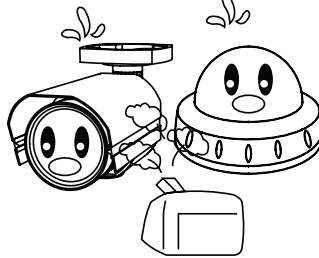
Precautions

Do not install the camera in extreme temperature conditions.



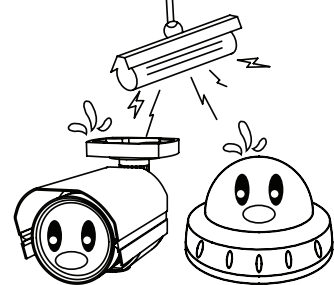
Only use the camera under conditions where temperatures are between -10°C and +50°C. Be especially careful to provide ventilation when operating under high temperatures.

Do not install or use the camera in an environment where the humidity is high.



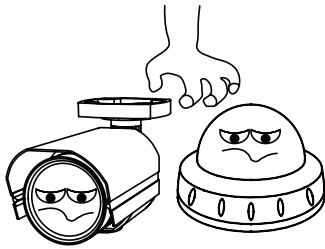
Do not install or use the camera in an environment where the humidity is high.

Do not install the camera under unstable lighting conditions.



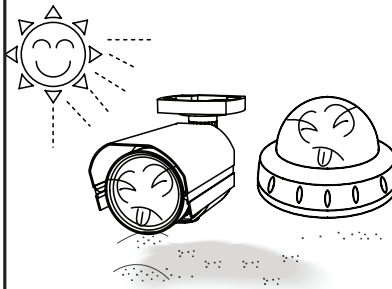
Severe lighting change or flicker can cause the camera to work improperly.

Do not touch the front lens of the camera.



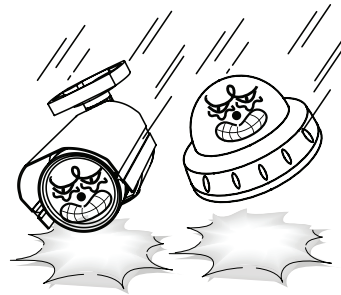
This is one of the most important parts of the camera. Be careful not to leave fingerprints on the lens cover.

Never keep the camera pointed directly at strong light.

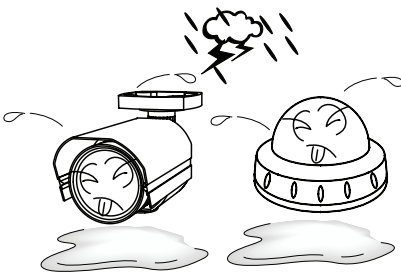


It can cause malfunctions to occur.

Do not drop the camera or subject it to physical shocks.

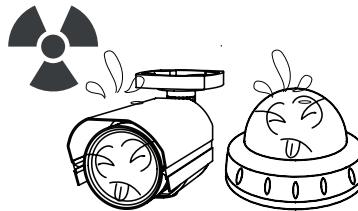


Do not expose internal grade cameras (Mini Domes) to rain or moisture.



If it gets wet, wipe dry immediately. Liquids can contain minerals that corrode the electronic components.

Do not expose the camera to radioactivity.



If exposed to radioactivity the CCD will fail.

NOTE

- If a camera is exposed to spotlights, or objects which reflect strong light, smearing or blooming may occur.
- Please check that the power satisfies the normal specification before connecting the camera.

Safety Instructions

Precautions for Use

- This camera should only be installed by qualified personnel.
- There are no user serviceable parts inside.
- Do not disassemble this camera other than to make initial adjustments.
- Use appropriate low voltage power cable to prevent fire or electrical shock.
- Please ensure your installation area can support the weight of the camera.

Please handle this Camera carefully

- Do not use strong or abrasive detergents when cleaning the camera.
- Do not expose the camera to direct sunlight.
- Do not install the camera in places where humidity is high.
- Do not install near cooling or heating devices.
- Do not expose an internal grade camera to rain or moisture.

How to Set Up the Camera Menu

Setup Menu

SET UP MENU	DEFAULT SET	MENU			
LENS	DC	BRIGHTNESS	0~100 Level Selectable		
		MODE	INDOOR		
		RETURN	RET SAVE & END		
	MANUAL				
EXPOSURE	SHUTTER	x2, x4, x6, x8, x10, x12, x14, x16, x24, x32, x64, x128, x256, AUTO,1/60,1/50, FLK, 1/250, 1/500, 1/1000, 1/2000, 1/5000, 1/10000, 1/20000, 1/50000 1/100000,			
	AGC	OFF			
		LOW			
		MIDDLE			
	SENSE-UP	AUTO	SENS-UP(x2 ~ x256 Level Selectable)		
			RETURN	RET SAVE & END	
		OFF			
	BLC	OFF			
		BLC	VALUE	LOW	
				MIDDLE	
			AREA	POSITION	
				SIZE	
			DEFAULT	AGAIN	
		RETURN	RET SAVE & END		
		HSBLC	GAIN	0 ~ 8 Level Selectable	
			MODE	ALL DAY	
			DEFAULT	NIGHT ONLY	
RETURN	RET SAVE & END				
D-WDR	OFF				
	INDOOR	LOW LEVEL	0 ~ 15 Level Selectable		
		HIGH LEVEL	0 ~ 15 Level Selectable		
		RETURN	RET SAVE&END		
	OUTDOOR	LOW LEVEL	0 ~ 15 Level Selectable		
		HIGH LEVEL	0 ~ 15 Level Selectable		
		RETURN	RET SAVE & END		
RETURN	RET SAVE & END				
WHITE BAL	ATW				
	AWB				
	AWC->SET				
	MANUAL	BLUE	0 ~100 Level Selectable		
RED		0 ~100 Level Selectable			
RETURN		RET SAVE & END			
DAY/NIGHT	COLOUR				
	B/W	BURST	ON/ OFF		
		IR SMART	VALUE	0 ~ 100 Level Selectable	
				POSITION	
			AREA	SIZE	
				RET/AGAIN	
		OFF			
	IR DWDR	OFF			
		1 ~ 15 Level Selectable			
	RETURN	RET SAVE & END			
	AUTO	DELAY	0 ~15 Level Selectable		
		D --> N (AGC)	0 ~ 255 Level Selectable		
		N --> D (AGC)	0 ~ 255 Level Selectable		
		RETURN	RET SAVE & END		
	EXT	START DELAY	0 ~15 Level Selectable		
END DELAY		0 ~15 Level Selectable			
RETURN		RET SAVE & END			

3DNR	OFF						
	ON	LEVEL	1 ~ 200 Level Selectable				
SPECIAL	CAM TITLE	RETURN	RET				
		SAVE & END					
	D-EFFECT	ON					
		OFF					
		FREEZE	ON/OFF				
			MIRROR	MIRROR			
				V-FLIP			
				ROTATE			
				OFF			
		D-ZOOM	ON	RANGE	x1 ~ x32 Selectable		
				PAN	-100 ~ 100 Selectable		
				TILT	-100 ~ 100 Selectable		
				RETURN	RET		
				SAVE & END			
		OFF					
		SMART DZOOM	ON	OFF			
				RANGE	x2 ~ x5 Selectable		
				POSITION	LEFT/RIGHT/UP/DOWN		
				SENSITIVITY	0 ~ 100 Level Selectable		
				TIME	0 ~ 15 Level Selectable		
				RETURN	RET		
				SAVE & END			
		DIS	ON/OFF				
		NEG. IMAGE	ON/OFF				
		RETURN	RET				
	SAVE & END						
	MOTION	ON	AREA	1 ~ 4 Selectable			
			AREA DISPLAY	ON	POSITION		
					SIZE		
					RET/AGAIN		
			OFF				
			VALUE	0 ~ 100 Level Selectable			
			MOTION VIEW	ON/OFF			
		RETURN	RET				
		SAVE & END					
	OFF						
	PRIVACY	ON	AREA	1 ~ 8 Selectable			
			AREA DISPLAY	ON	TOP LEFT		
					TOP RIGHT		
					BTM LEFT		
					BTM RIGHT		
					POSITION		
					RET/AGAIN		
					OFF		
			COLOUR	1 ~ 15 Selectable			
		TRANSPAR	0 ~ 3 Selectable				
		RETURN	RET				
	SAVE & END						
	OFF						
	DEFECT	SENS UP	x4 ~ x128 Level Selectabl				
		DIFF	0 ~ 3 Level Selectable				
		THRESHOLD	0 ~ 4 Level Selectable				
		START					
		RETURN	RET				
	SAVE & END						
	DEFOG	OFF					
		MANUAL	LEVEL	0 ~ 31 Level Selectable			
			COLOUR GAIN	0 ~ 10 Level Selectable			
			EDGE GAIN	0 ~ 10 Level Selectable			
			GAMMA	USER, 0.05 ~ 1.00 Level Selectable			
RETURN			RET				
SAVE & END							
AUTO		DETECT LEVE	0 ~ 5 Level Selectable				
	RETURN	RET					
SAVE & END							
RETURN	RET						
END							

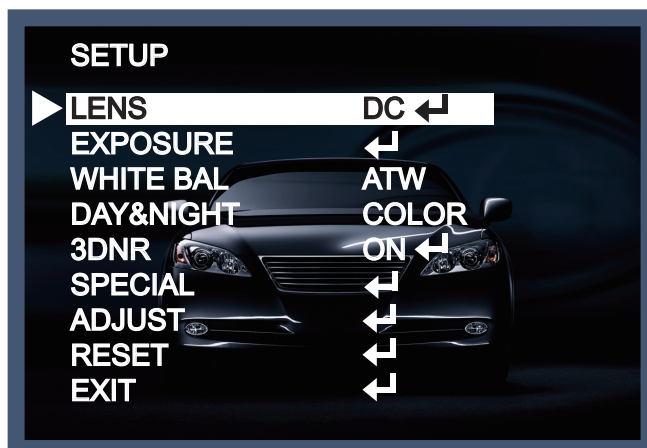
ADJUST	SHARPNESS	0 ~ 31 Level Selectable			
	MONITOR	LCD	GAMMA	USER, 0.05 ~ 1.00 Level Selectable	
			LEVEL	0 ~ 63 Level Selectable	
			BLUE GAIN	0 ~ 255 Level Selectable	
			RED GAIN	0 ~ 255 Level Selectable	
			RETURN	RET	
				SAVE & END	
		CRT	LEVEL	0 ~ 63 Level Selectable	
			BLUE GAIN	0 ~ 255 Level Selectable	
			RED GAIN	0 ~ 255 Level Selectable	
			RETURN	RET	
				SAVE & END	
		USER	GAMMA	USER, 0.05 ~ 1.00 Level Selectable	
			LEVEL	0 ~ 63 Level Selectable	
			BLUE GAIN	0 ~ 255 Level Selectable	
			RED GAIN	0 ~ 255 Level Selectable	
			RETURN	RET	
				SAVE & END	
RESET	FACTORY	RESET			
	RETURN	RET			
		SAVE & END			
EXIT					

Menu Set Up

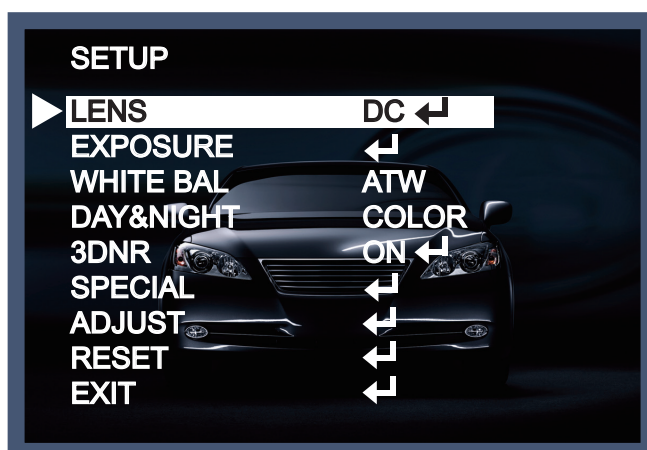
Menu settings can be selected using the buttons inside the camera.

1. Press the SET button.

* The Set Up menu will be displayed on the monitor.



2. Move and select the required function using the Up and Down buttons.



3. Change menu settings using the Left or Right buttons.

* Available values are displayed by pressing the Left and Right buttons.

Press the buttons until the desired value is displayed.

4. After altering the setting, move the cursor to EXIT and press the SET button to exit.

NOTE

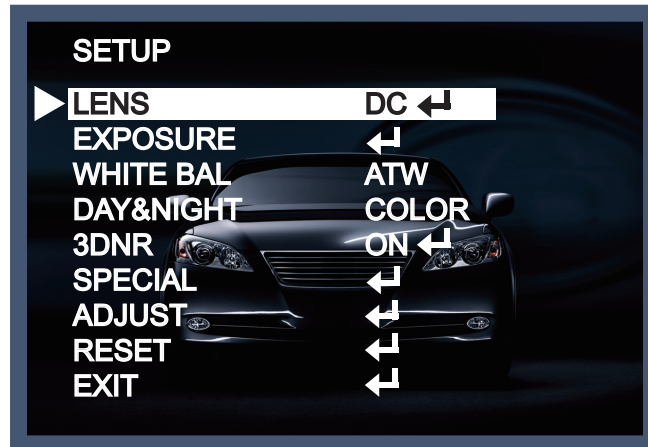
* Move to the available submenu by moving  arrow to desired feature.

* A submenu is not available when this symbol is displayed “---”.

Lens

This function adjusts the brightness of the image being displayed on the monitor.

1. Move the cursor to LENS using the Up and Down buttons on the Set Up menu screen.
2. Select the desired lens type by pressing the Left or Right button.



NOTE

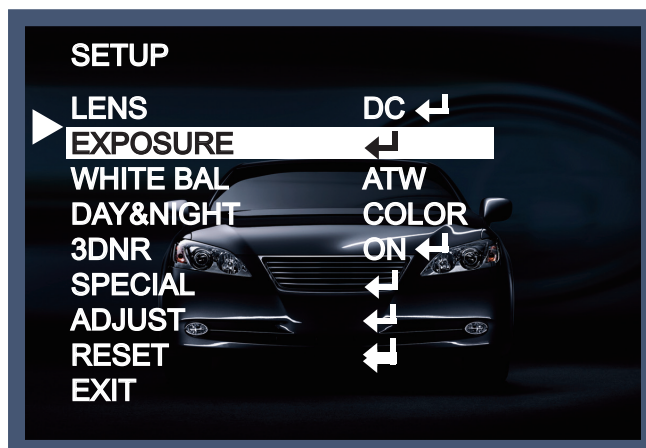
* When DC is selected, the brightness control range is 1~100.



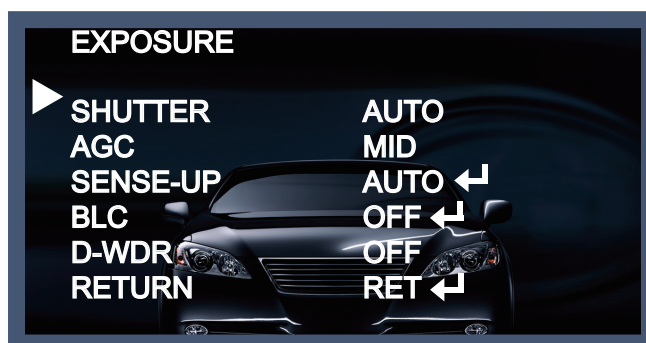
* When you select the mode setup, you can select INDOOR or OUTDOOR depending on the installation environment.

Exposure

1. On the Set Up menu, Select EXPOSURE using the Up or Down buttons.



2. Select the desired shutter mode by pressing the Left or Right button.



- Electronic Shutter: DC Lens is set by default to 1/50 and Manual is defaulted to the Auto shutter setting, but can be adjusted from 1/50 to 1/100,000. Sens Up does not work when in Manual.
 - DIS (Digital Image Stabiliser): Select DIS mode if the camera is moving due to an environmental condition. It stabilises the image. Sens Up does not work when in DIS mode.
- AGC (Auto Gain Control): The higher the AGC level is, the more noise will appear.
 - Off: Deactivated
 - Low: 28dB Gain is applied.
 - Mid: 32dB Gain is applied.
 - High: 36dB Gain is applied.

NOTE

* Sens Up mode does not work while the Electronic shutter is set to Manual or DIS.

- Sens Up: The degree of darkness is automatically sensed and compensated for.
 - Off: Deactivated status
 - Auto: Sens Up is in automatic mode and the gain is decided by the low luminance level.



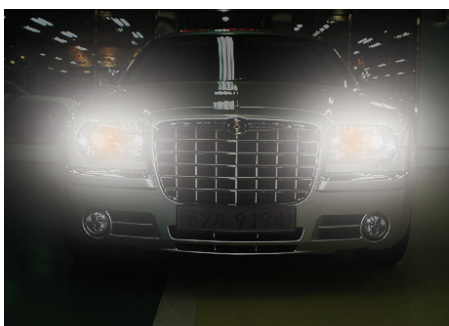
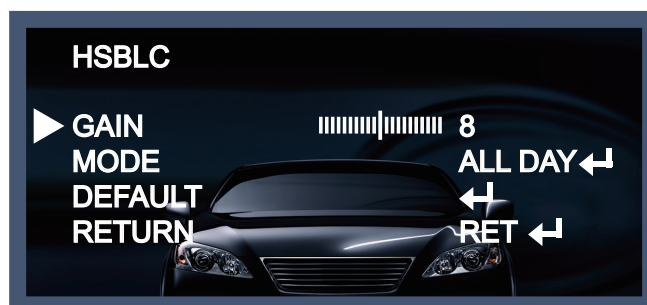
NOTE

* Sens Up mode does not work when AGC is off.

- BLC (Black Light Compensation): The camera is equipped with 3D-DNR which provides intelligent light level control to overcome strong backlit conditions.
 - Off: Deactivated status
 - BLC: The BLC area can be set by the user to compensate for backlit installation environments.



- HSBLC: If the scene contains an extremely bright light source, such as car headlights, HSBLC can be used to mask them. This is especially useful for number plate recognition.



Normal



HSBLC

- D-WDR: This camera has 3D-DNR which provides intelligent light level control even when strong backlight is present.
 - Off: Deactivated status.
 - Indoor Compensation: To be used when the camera is installed internally.
 - Outdoor Compensation: To be used when the camera is installed externally.



WDR OFF



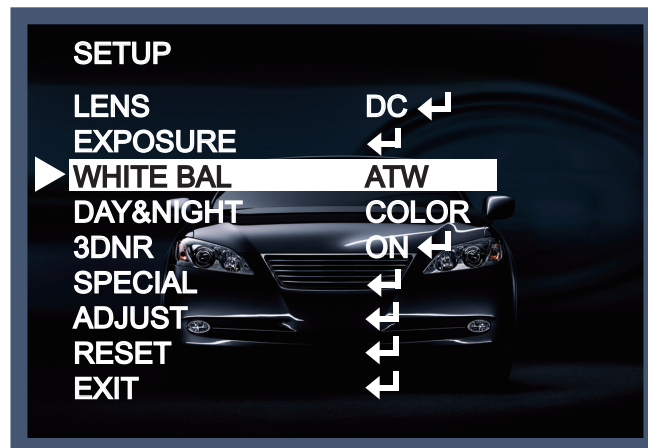
WDR ON

- Return: Save changes and return to the previous menu.

White Balance

Use this function when adjustment of the colour output is needed.

1. Move the triangular cursor to WHITE BAL on the Set Up menus screen, using the Up and Down buttons
2. Select the desired mode using the Left and Right buttons.



Select the most appropriate mode for the prevailing installation conditions.

- ATW: Use this mode when the colour temperature is from 1,800K to 10,500K.
- AWB: Use this mode when the colour temperature is from 3,000K to 7,000K.
- AWC → SET: Direct and focus the camera toward a blank white sheet of paper and press the SET button. If the lighting conditions change re-calibration will be required.
- Manual: Manual setting allows for more detailed colour control. First adjust the white balance using either ATW or AWB. Then change to manual mode and press the SET button.



NOTE

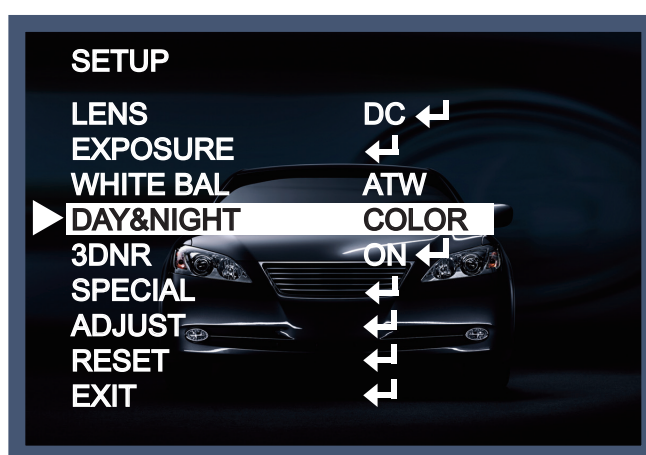
White Balance may not work correctly in the following conditions:

- * When there is a very high colour temperature within the subject, for example a sunset in a clear sky.
- * It is very dark
- * If the camera is pointed towards fluorescent light, or there is a drastic change in lighting, the White Balance operation may become unstable.
- * In these conditions, AWB mode is recommended.

Day & Night

The Day & Night menu is used to set the camera video output.

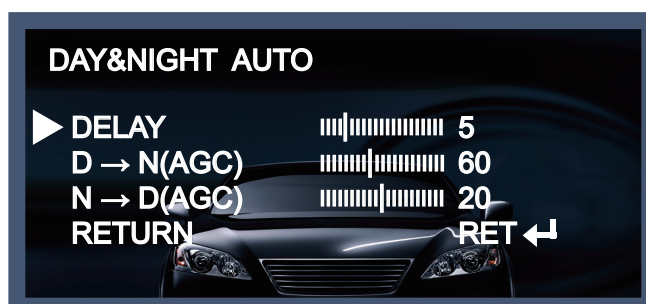
1. Select Day & Night using the Up or Down button on the set up menu screen.
2. Select the desired mode using the Left or Right buttons.



- Colour: The video output remains in colour mode at all times.
- B/W: The video output remains in monochrome at all times.
 - Burst: Removes the colour element and makes the video output monochrome.
 - Smart IR: Eliminates IR saturation caused by objects which are close to the camera. Only for cameras equipped or used with IR LED's.



- Auto: Automatically switches to colour during daytime and monochrome at night time, or low light environments. The switching time is adjustable - both day to night and night to day.

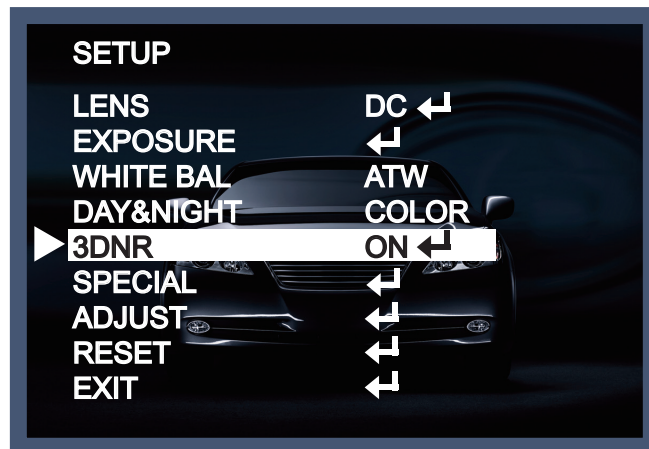


- EXT: Select the mode using the Right or Left button.

3DNR

3DNR decreases noise caused by low luminance levels.

1. Move the cursor to 3DNR using the Up and Down buttons.
2. Select the mode using the Right or Left button.



3. The 3DNR level is adjustable from 0~200. Change to On and press the SET button.

- Off: Deactivated status.
- On: The compensation level is 0~200.



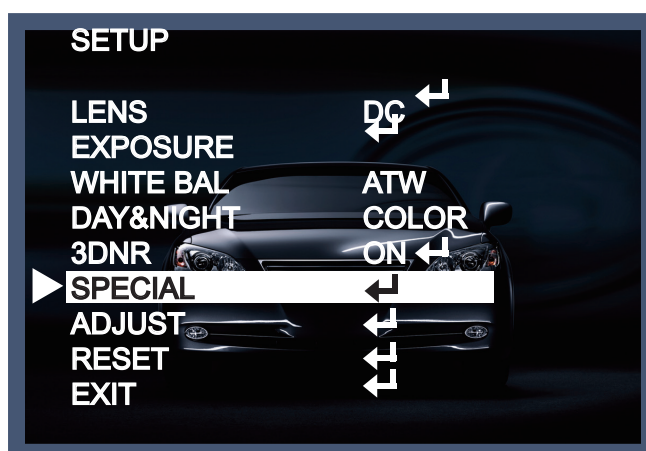
NOTE

* When the 3DNR function is activated noise cancellation is extremely effective. However, ghosting can occur if the level is set too high.

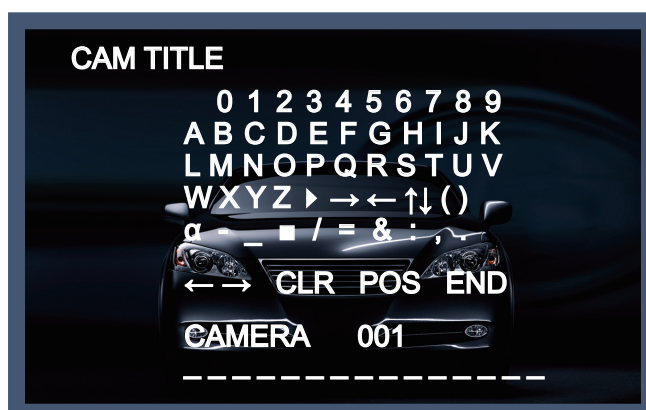
Special

This menu is used to select additional control functions.

1. When the SETUP menu is displayed, select SPECIAL using the Up and Down buttons.
2. Select one of the modes using the Up and Down buttons.



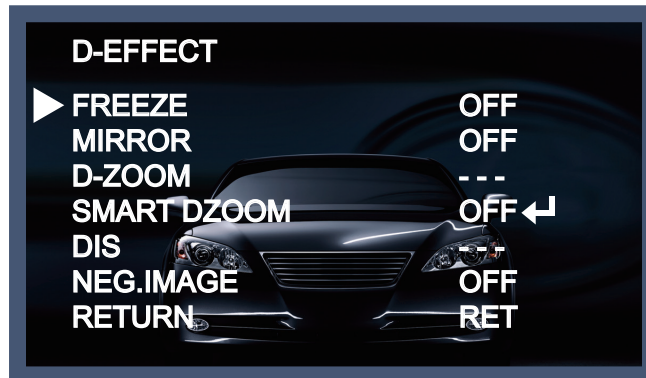
- CAM (Camera) TITLE: This function is used to display the camera title on the monitor and recorded footage. Up to 15 alphanumeric characters can be displayed, and positioned, on the screen. Move the cursor to CAMERA TITLE using the Up and Down buttons and set to On using the left or Right button. Enter the chosen ID and press the Set button on END to complete the procedure.



NOTE

* If you need to erase a character, move the cursor on the empty space before the 0. Characters will be erased, one by one, by pressing the Set button. Selecting the CLR option deletes all characters.

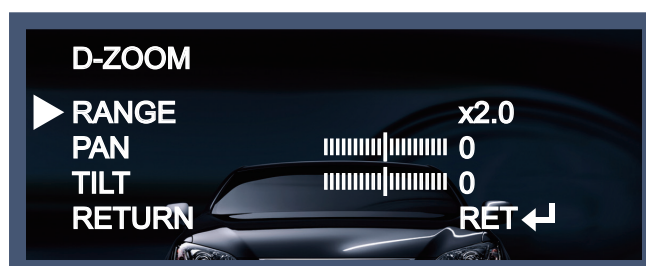
- D-EFFECT: This function allows the video image to be manipulated.



- FREEZE: Causes the video image to freeze.
- MIRROR: Causes the video image to be reversed vertically or horizontally.



- D-Zoom: Implements a digital zoom of the image with x1~x32 magnification.



NOTE

* When you use the digital zoom, SMART DZOOM and DIS cannot be used.

- **SMART DZOOM:** This function is interlocked with D-ZOOM to distinguish specific objects in greater detail when movement is detected in programmed areas. This function is much more effective when used with optical zoom.



- **DIS (Digital Image Stabiliser):** DIS compensates for slight image movements in the camera moves because of environmental conditions i.e. wind.



DIS OFF

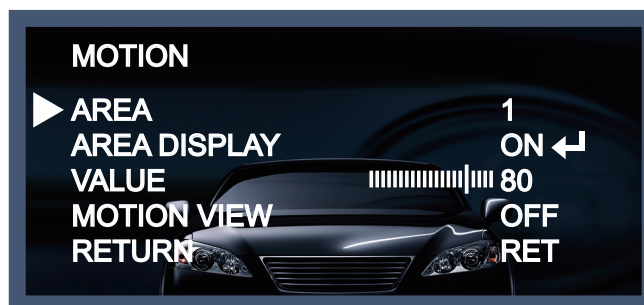


DIS ON

- **NEG IMAGE**
 - ON: Output a negative image.
 - OFF: Normal image

- Return: Save and return to the SPECIAL menu.

- **MOTION:** Detects movement of a subject within a pre-defined area(s). When MOTION is On, a new sub-menu is displayed when the Set button is pressed.



- Area: Up to 4 areas can be designated.
- AREA DISPLAY: The area screen should be activated - set to On - after selecting it. Activation displays the selected area on the screen.
- VALUE (Sensitivity): The higher the set value, the higher the sensitivity.
- Motion View: If the camera detects motion, the sentence "Motion Detected" appears on the monitor.
- Return: Save and return to the SPECIAL menu.

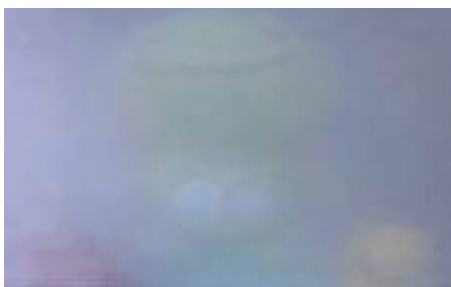
- **PRIVACY:** The Privacy function is used to mask certain areas of the surveillance scene. Up to 8 areas can be defined and their size is adjustable. Fifteen colours are available to differentiate areas. When RETURN is selected all changes are saved.



- **DEFECT:** White spot defects can be eliminated - up to a maximum of 512 points.



- Sens Up: To eliminate defects set Sens Up to a high level. We recommend using a maximum value of x32.
 - DIFF: You can set the size of DEFECT from 0 to 3.
 - THRESHOLD: The higher the level is, the more defects appear.
 - START: Press the SET button on the "<" symbol and the defect elimination process begins. The message "Processing Now" is displayed on the monitor.
 - Return: Save the setting value of this menu, return to the SPECIAL menu.
- **DEFOG:** The Defog function aids object recognition in foggy conditions.



DEFOG OFF



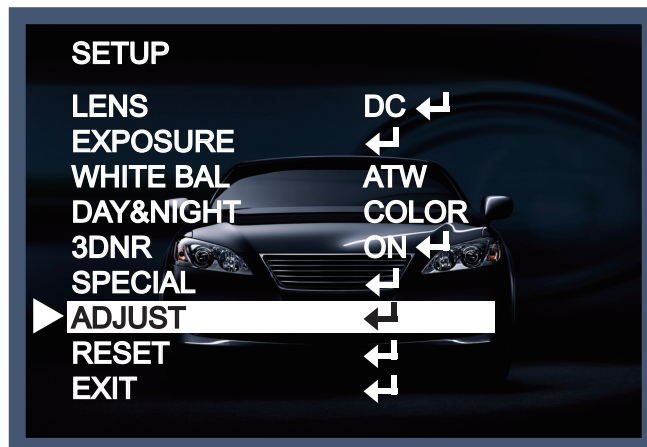
DEFOG ON

- Return: Save the set values and exit from the menu. Adjust

Adjust

This function is used to adjust the video output, and to select the type of monitor where it's being displayed.

1. When the Set Up menu is displayed select ADJUST using the Up and Down buttons.



- Sharpness: The outline of an image becomes sharp and more distinct as the level of sharpness increases.
- Monior: Select the appropriate monitor type.
 - LCD: Select this option when using a LCD monitor.
(You can adjust the Gamma, Level, Blue & Red Gain on the submenu)
 - CRT: Select this option when using a CRT monitor.
(You can adjust the Level, Blue & Red Gain on the submenu)
 - USER: Select this option if the monitor being used has a poor contrast rate. Go into the submenu and adjust the Gamma, Level and Red and Blue Gain.

Reset

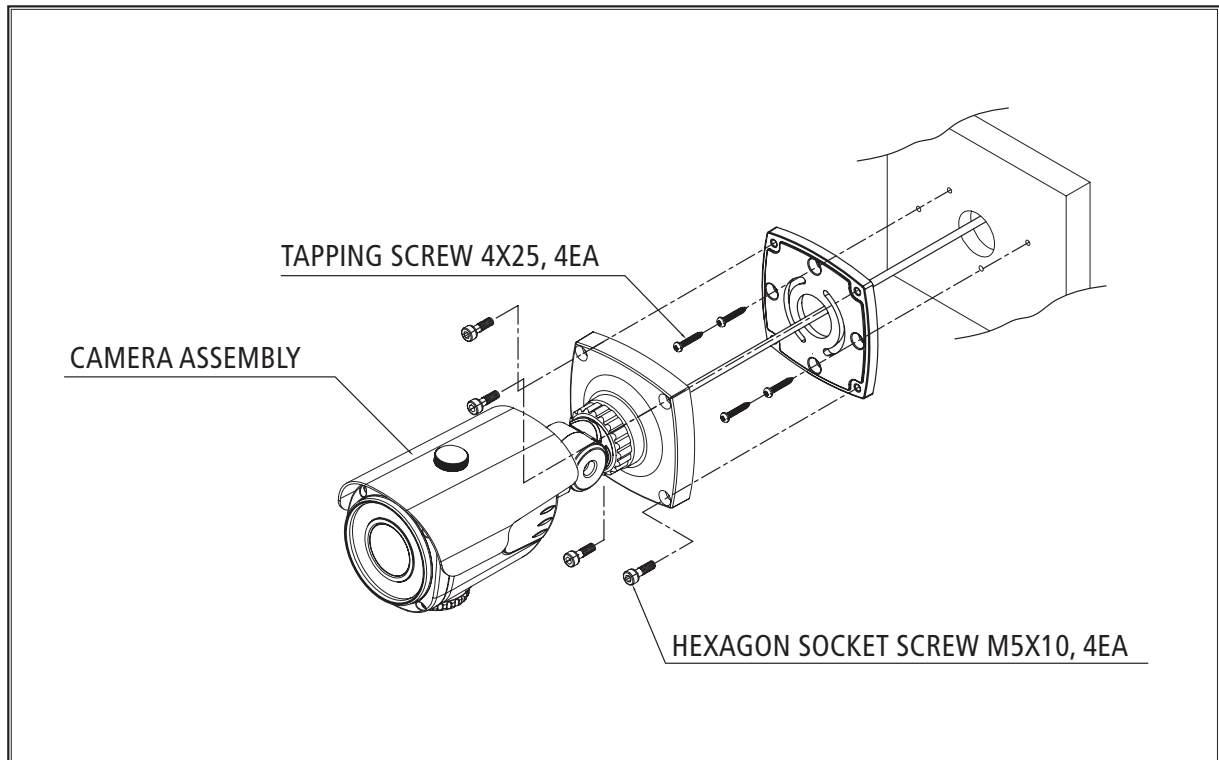
The camera is reset to factory default values.

Exit

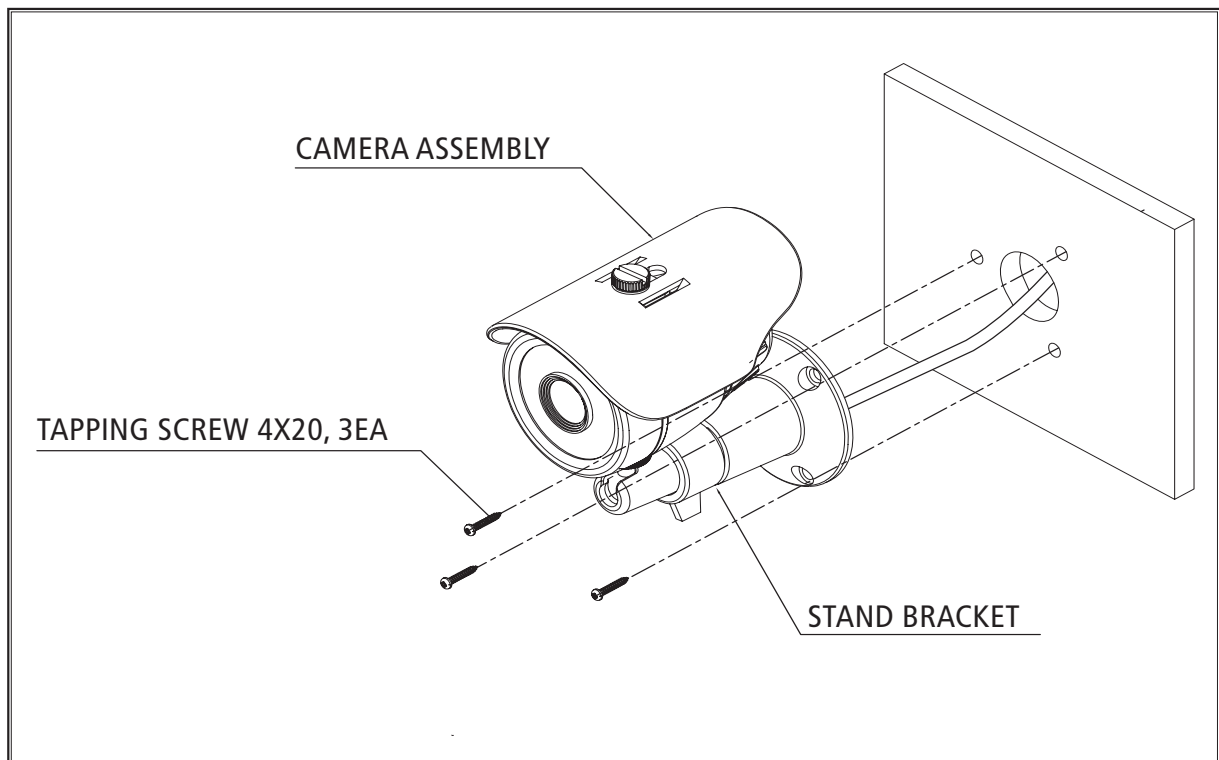
Select EXIT and press the Set button to save the current settings and exit the Set Up menu.

Camera Installation

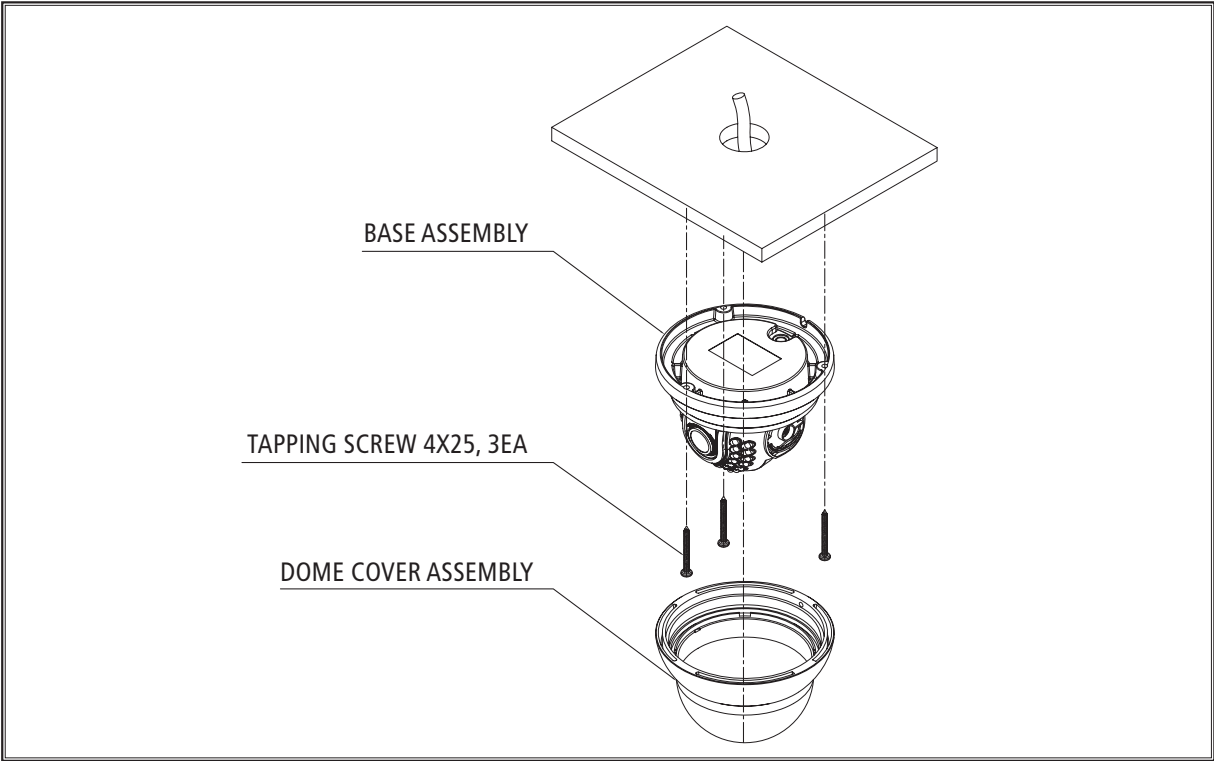
1. W96BVIR



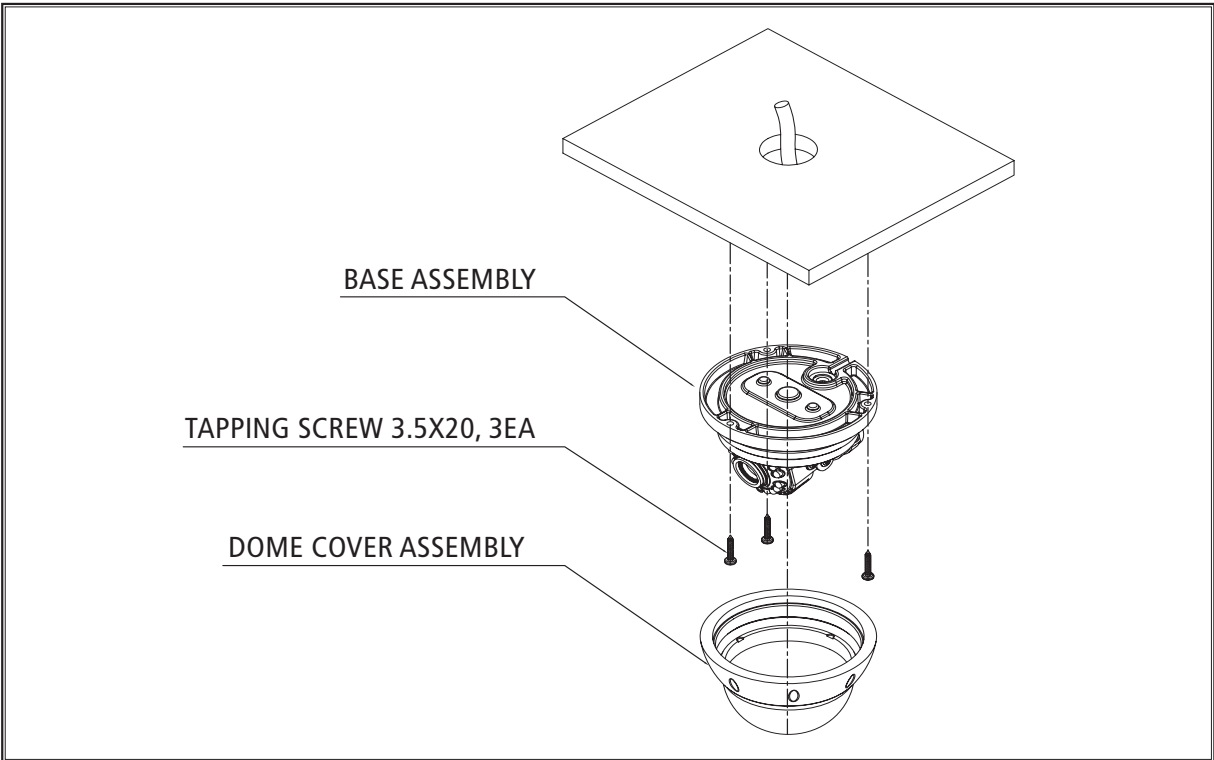
2. W96BIR



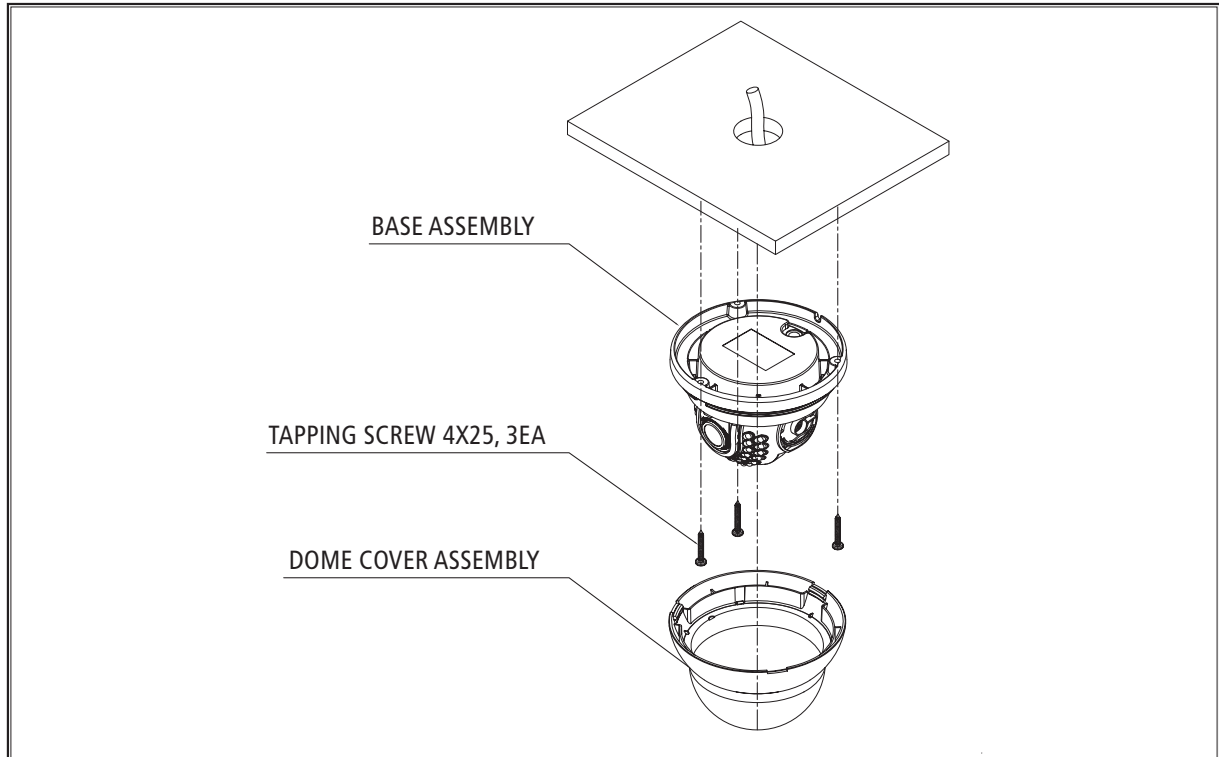
3. W96VDVIR, W96VDV



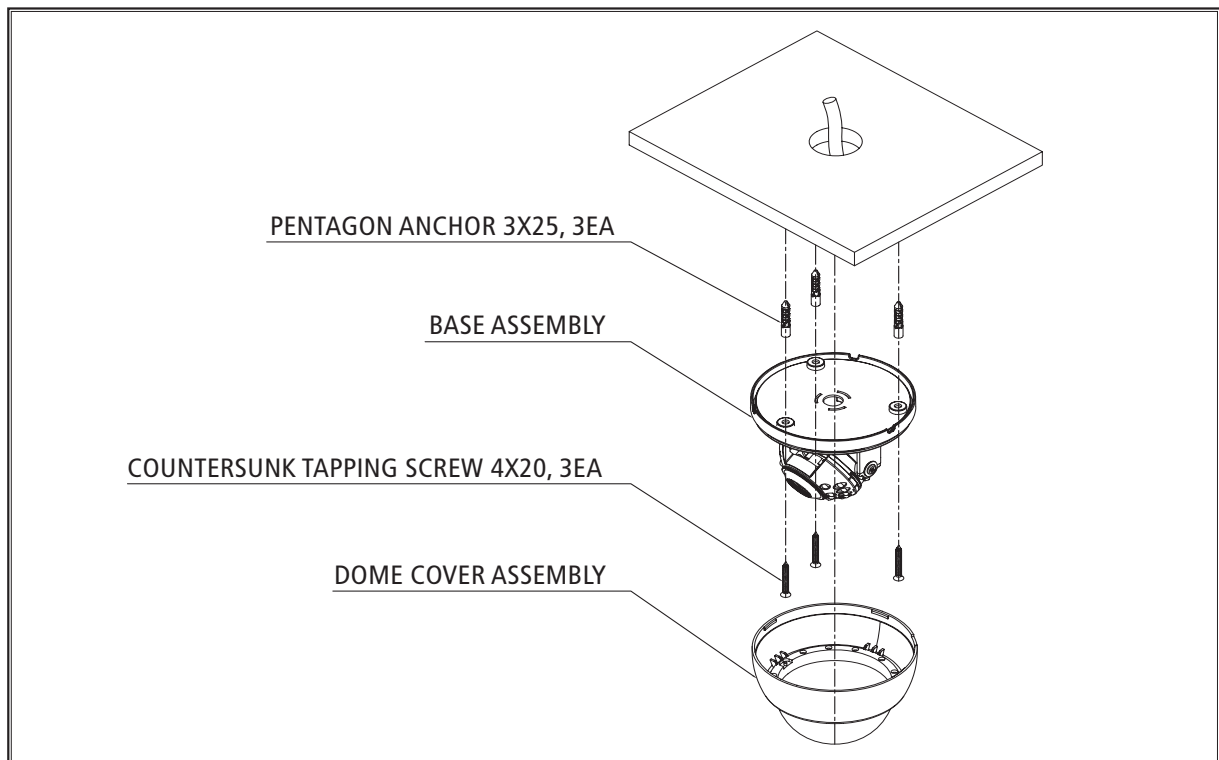
4. W96VD



5. W96MDVIR(W), W96MDV(W)



6. W96MDW



Specification



W96BVIR

Model Number		W96BVIR
Image Sensor	Sensor Type	1/3" Colour 960H CCD
	Effective Pixels	976 (H) x 494 (V) approx. 480K Pixels (NTSC) 976 (H) x 582 (V) approx. 570K Pixels (PAL)
	Total Pixels	1020 (H) x 508 (V) approx. 520K Pixels (NTSC) 1020 (H) x 596 (V) approx. 610K Pixels (PAL)
Resolution		Colour: 650TVL, B&W: 700TVL
Scanning System		2:1 Interlaced
Minimum Illumination		IR LED On: 0 Lux @F1.2 (Sens Up x256)
Video Output		1.0Vp-p Composite, 75 Ohms
S/N Ratio		52dB
Lens	Focal Length	f = 2.8mm ~ 12mm
	Drive Type	DC Auto Iris
Function	AGC	Low / Middle / High
	Sens Up	OFF / AUTO (x2 ~ x256)
	Backlight Compensation	OFF / BLC / HLC
	D-WDR	OFF / INDOOR / OUTDOOR
	Motion Detection	4 Zones
	Privacy Masking	8 Zones
	DIS (Digital Image Stabiliser)	OFF / ON
	Day & Night	Auto (AGC) / Day / Night / Ext
	White Balance	ATW / AWB / AWC -> SET / Manual (1,800K ~ 11,000K)
	3DNR (2D or 3D)	1 ~ 200 Level Selectable
	Mirror	H-Rev / V-Rev / Rotate / PIP
	D-Zoom	x32
	Smart DZoom (Motion Detection)	OFF / ON
	Defog	OFF / MANUAL / AUTO
	Blemish Compensation	Auto (512 Points)
Power Source		DC12V ±10%
Power Consumption (IR's On / Max.)		DC: 640mA ±10% (Max.)
IR Spectrum		850nm/30° x ø5 x 42EA
Viewable Distance		Up to 25m in Ideal Conditions
Storage Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH
Operating Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH)
Dimensions (W x H x D)		90 x 95 x 210mm
Weight		1400g

Specification



Model Number		W96BIR
Image Sensor	Sensor Type	1/3" Colour 960H CCD
	Effective Pixels	976 (H) x 494 (V) approx. 480K Pixels (NTSC) 976 (H) x 582 (V) approx. 570K Pixels (PAL)
	Total Pixels	1020 (H) x 508 (V) approx. 520K Pixels (NTSC) 1020 (H) x 596 (V) approx. 610K Pixels (PAL)
Resolution		Colour: 650TVL, B&W: 700TVL
Scanning System		2:1 Interlaced
Minimum Illumination		IR LED On: 0 Lux @F1.2 (Sens Up x256)
Video Output		1.0Vp-p Composite, 75 Ohms
S/N Ratio		52dB
Lens		f = 3.6mm Fixed
Power Source		DC12V $\pm$ 10%
Power Consumption (IR's On / Max.)		DC: 450mA $\pm$ 10% (Max.)
IR Spectrum		850nm/30° x ϕ 5 x 24EA
Viewable Distance		Up to 15m in Ideal Conditions
Storage Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH
Operating Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH
Dimensions (W x H x D)		66 x 67 x 77mm
Weight		700g

Specification



W96VDVIR

W96VDV

Model Number		W96VDVIR	W96VDV
Image Sensor	Sensor Type	1/3" Colour 960H CCD	
	Effective Pixels	976 (H) x 494 (V) approx. 480K Pixels (NTSC) 976 (H) x 582 (V) approx. 570K Pixels (PAL)	
	Total Pixels	1020 (H) x 508 (V) approx. 520K Pixels (NTSC) 1020 (H) x 596 (V) approx. 610K Pixels (PAL)	
Resolution		Colour: 650TVL, B&W: 700TVL	
Scanning System		2:1 Interlaced	
Minimum Illumination		IR LED ON: 0 Lux @F1.2 (Sens Up x256)	Colour: 0.01 Lux @F1.2 (Sens Up x256)
			B&W: 0.001 Lux @F1.2 (Sens Up x256)
Video Output		1.0Vp-p Composite. 75 Ohms	
S/N Ratio		52dB	
Lens	Focal Length	f = 2.8mm ~ 12mm	
	Drive Type	DC Auto Iris	
Function	AGC	Low / Middle / High	
	Sens Up	OFF / Auto (x2 ~ x256)	
	Backlight Compensation	OFF / BLC / HLC	
	D-WDR	OFF / INDOOR / OUTDOOR	
	Motion Detection	4 Zones	
	Privacy Masking	8 Zones	
	DIS (Digital Image Stabiliser)	OFF / ON	
	Day & Night	Auto (AGC) / Day / Night / Ext	
	White Balance	ATW / AWB / AWC - > SET / Manual (1,800K ~ 11,000K)	
	3DNR (2D or 3D)	1 ~ 200 Level Selectable	
	Mirror	H-Rev / V-Rev / Rotate / PIP	
	D-Zoom	x32	
	Smart DZoom (Motion Detection)	OFF / ON	
	Defog	OFF / MANUAL / AUTO	
	Blemish Compensation	Auto (512 Points)	
Power Source		DC12V ±10%	
Power Consumption (IR's On / Max.)		DC: 400mA ±10%	DC: 200mA ±10%
IR Spectrum		850nm/30° x ø5 x 28EA	
Viewable Distance (Ideal Conditions)		Indoor Up to 30m	
Storage Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH	
Operating Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH	
Dimensions (W x H x D)		130 x 100 x 130mm	
Weight		900g	

Specification



W96VD

Model Number		W96VD
Image Sensor	Sensor Type	1/3" Colour 960H CCD
	Effective Pixels	976 (H) x 494 (V) approx. 480K Pixels (NTSC) 976 (H) x 582 (V) approx. 570K Pixels (PAL)
	Total Pixels	1020 (H) x 508 (V) approx. 520K Pixels (NTSC) 1020 (H) x 596 (V) approx. 610K Pixels (PAL)
Resolution		Colour: 650TVL, B&W: 700TVL
Scanning System		2:1 Interlaced
Minimum Illumination		Colour: 0.01 Lux @ F1.2 (Sens Up x256) B&W: 0.001 Lux @ F1.2 (Sens Up x256)
Video Output		1.0Vp-p Composite, 75 Ohms
S/N Ratio		52dB
Lens	Focal Length	f = 3.6mm Fixed
Function	AGC	Low / Middle / High
	Sens Up	OFF / AUTO (x2 ~ x256)
	Backlight Compensation	OFF / BLC / HLC
	D-WDR	OFF / INDOOR / OUTDOOR
	Motion Detection	4 Zones
	Privacy Masking	8 Zones
	DIS (Digital Image Stabiliser)	OFF / ON
	Day & Night	Auto (AGC) / Day / Night / Ext
	White Balance	ATW / AWB / AWC -> SET / Manual (1,800K ~ 11,000K)
	3DNR (2D or 3D)	1 ~ 200 Level Selectable
	Mirror	H-Rev / V-Rev / Rotate / PIP
	D-Zoom	x32
	Smart DZoom (Motion Detection)	OFF / ON
	Defog	OFF / MANUAL / AUTO
	Blemish Compensation	Auto (512 Points)
Power Source		DC12V $\pm$ 10%
Power Consumption		DC: 200mA $\pm$ 10%
Storage Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH
Operating Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH)
Dimensions (W x H x D)		110 x 82 x 110mm
Weight		700g

Specification



Model Number		W96MDVIR / W96MDVIRW	W96MDV / W96MDVW
Image Sensor	Sensor Type	1/3" Colour 960H CCD	
	Effective Pixels	976 (H) x 494 (V) approx. 480K Pixels (NTSC) 976 (H) x 582 (V) approx. 570K Pixels (PAL)	
	Total Pixels	1020 (H) x 508 (V) approx. 520K Pixels (NTSC) 1020 (H) x 596 (V) approx. 610K Pixels (PAL)	
Resolution		Colour: 650TVL, B&W: 700TVL	
Scanning System		2:1 Interlaced	
Minimum Illumination		IR LED ON: 0 Lux @F1.2 (Sens Up x256)	Colour: 0.01 Lux @F1.2 (Sens Up x256)
			B&W: 0.001 Lux @F1.2 (Sens Up x256)
Video Output		1.0Vp-p Composite. 75 Ohms	
S/N Ratio		52dB	
Lens	Focal Length	f = 2.8mm ~ 12mm	
	Drive Type	DC Auto Iris	
Function	AGC	Low / Middle / High	
	Sens Up	OFF / Auto (x2 ~ x256)	
	Backlight Compensation	OFF / BLC / HLC	
	D-WDR	OFF / INDOOR / OUTDOOR	
	Motion Detection	4 Zones	
	Privacy Masking	8 Zones	
	DIS (Digital Image Stabiliser)	OFF / ON	
	Day & Night	Auto (AGC) / Day / Night / Ext	
	White Balance	ATW / AWB / AWC - > SET / Manual (1,800K ~ 11,000K)	
	3DNR (2D or 3D)	1 ~ 200 Level Selectable	
	Mirror	H-Rev / V-Rev / Rotate / PIP	
	D-Zoom	x32	
	Smart DZoom (Motion Detection)	OFF / ON	
	Defog	OFF / MANUAL / AUTO	
	Blemish Compensation	Auto (512 Points)	
Power Source		DC12V ±10%	
Power Consumption (IR's On / Max.)		DC: 400mA ±10%	DC: 200mA ±10%
IR Spectrum		850nm/30° x ø5 x 28EA	
Viewable Distance		Up to 25m in Ideal Conditions	
Storage Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH	
Operating Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH	
Dimensions (W x H x D)		130 x 100 x 130mm	
Weight		600g	

Specification

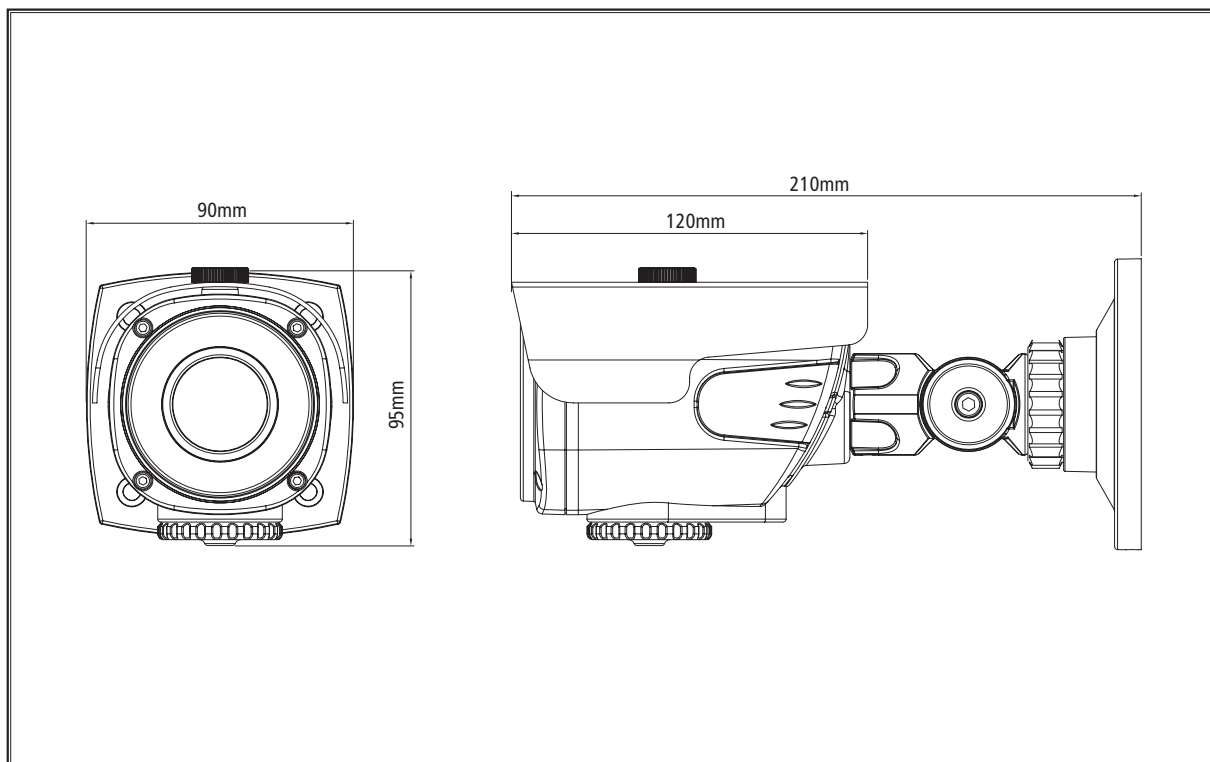


W96MDW

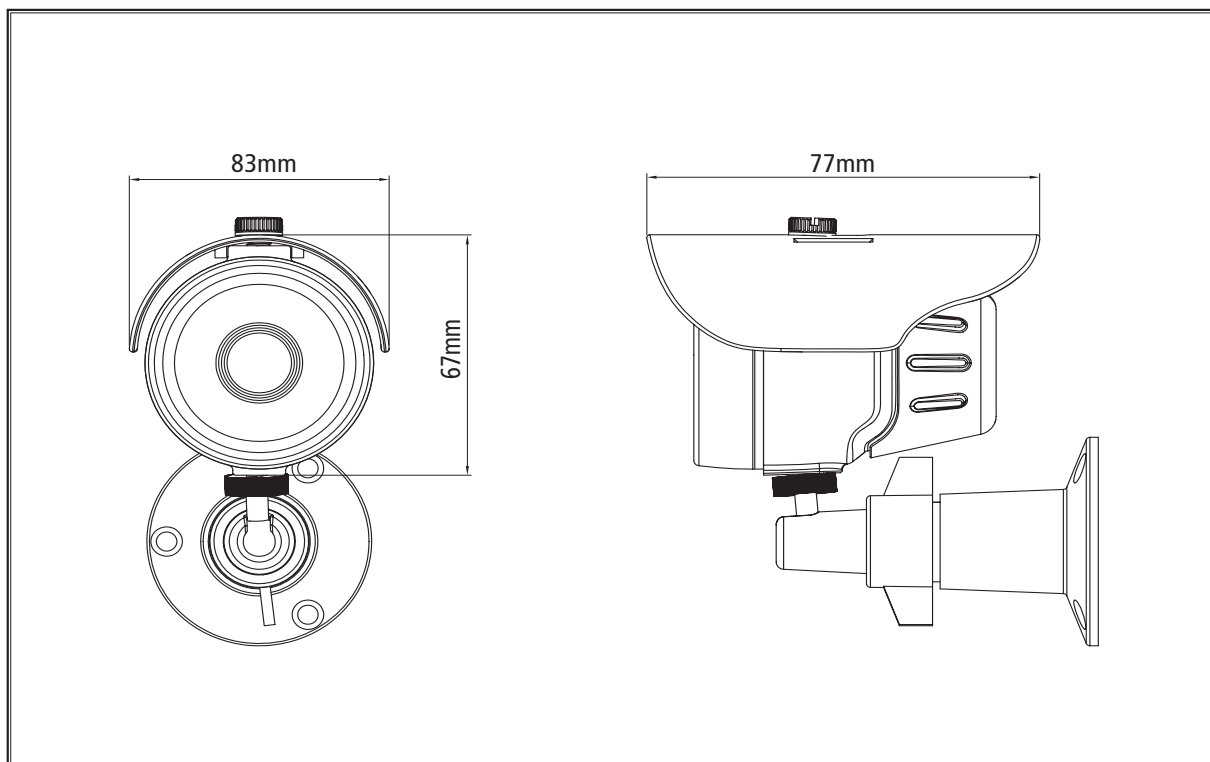
Model Number		W96MDW
Image Sensor	Sensor Type	1/3" Colour 960H CCD
	Effective Pixels	976 (H) x 494 (V) approx. 480K Pixels (NTSC) 976 (H) x 582 (V) approx. 570K Pixels (PAL)
	Total Pixels	1020 (H) x 508 (V) approx. 520K Pixels (NTSC) 1020 (H) x 596 (V) approx. 610K Pixels (PAL)
Resolution		Colour: 650TVL, B&W: 700TVL
Scanning System		2:1 Interlaced
Minimum Illumination		Colour: 0.01 Lux @F1.2 (Sens Up x256) B&W: 0.001 Lux @F1.2 (Sens Up x256)
Video Output		1.0Vp-p Composite, 75 Ohms
S/N Ratio		52dB
Lens		f = 3.6mm Fixed
Power Source		DC12V $\pm$ 10%
Power Consumption		DC: 200mA $\pm$ 10%
IR Spectrum		850nm/30° x ϕ 5 x 24EA
Viewable Distance		Up to 15m in Ideal Conditions
Storage Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH
Operating Conditions		Temperature: -20°C ~ +60°C Humidity: 20%RH ~ 95%RH)
Dimensions (W x H x D)		110 x 85 x 110mm
Weight		300g

Camera Dimensions

1. W96BVIR

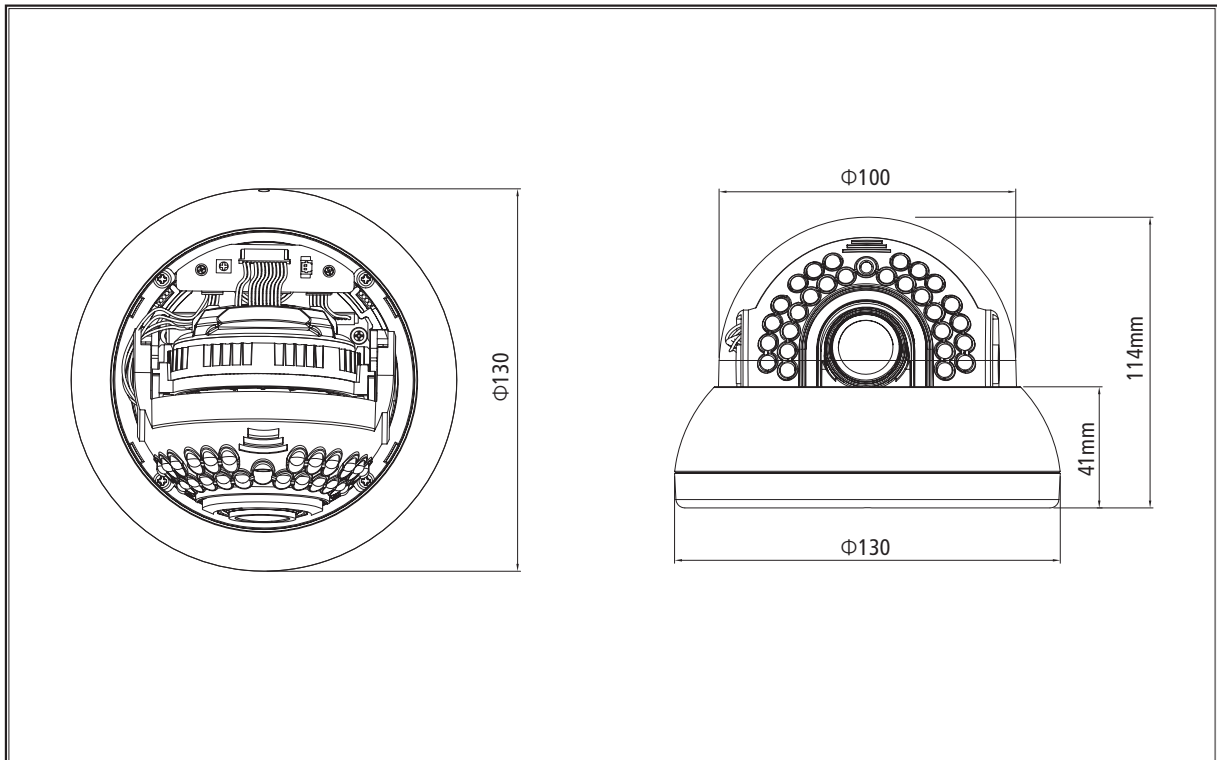


2. W96BIR

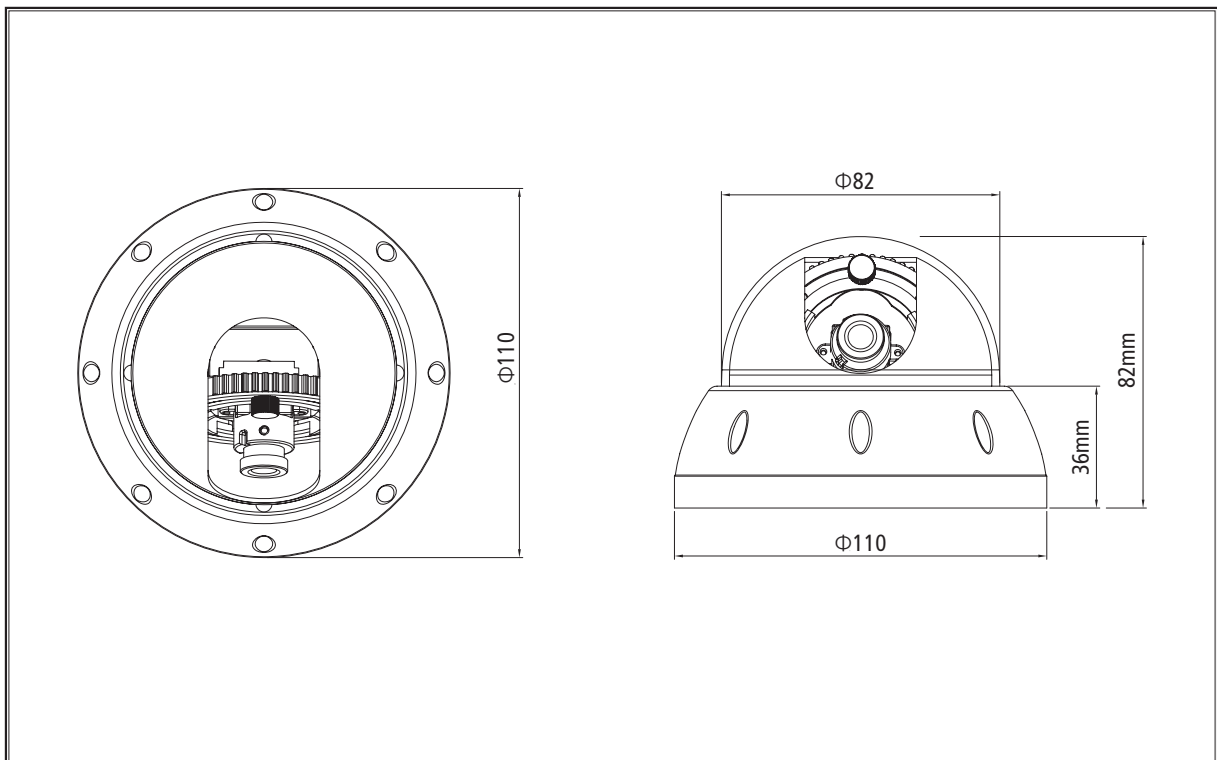


Camera Dimensions

3. W96VDVIR, W96VDV

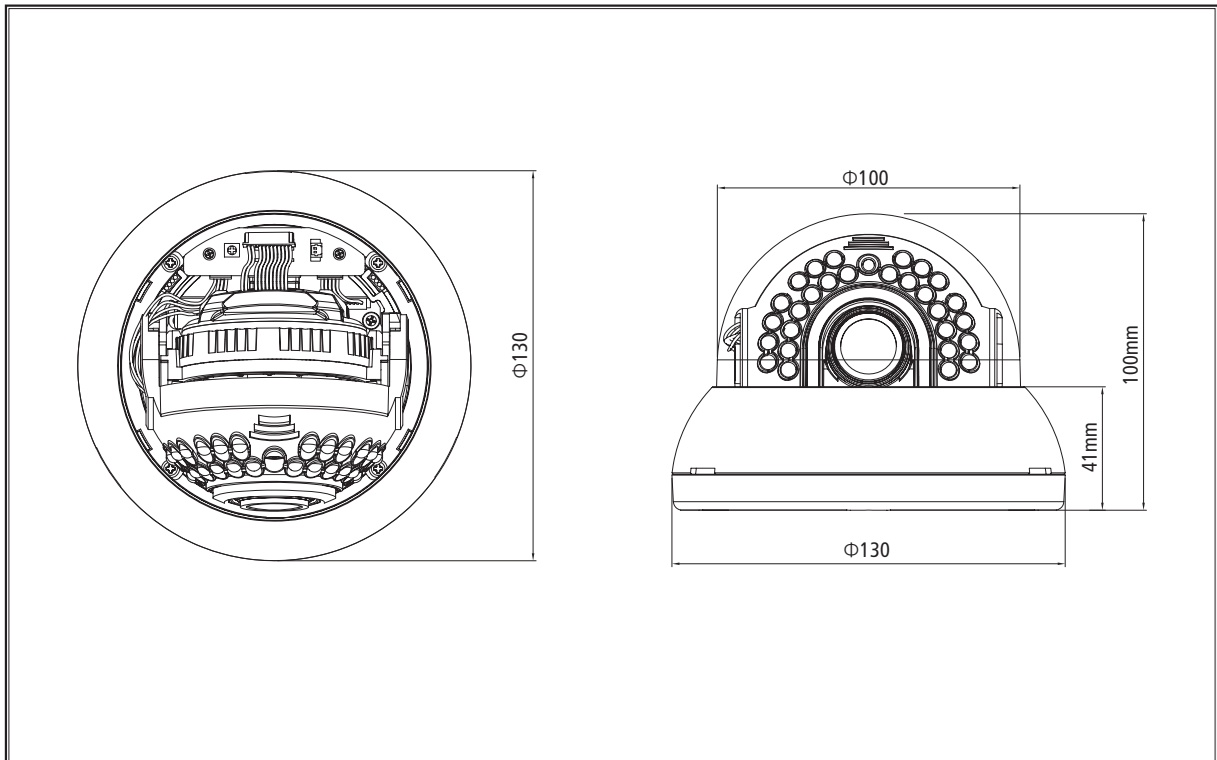


4. W96VD

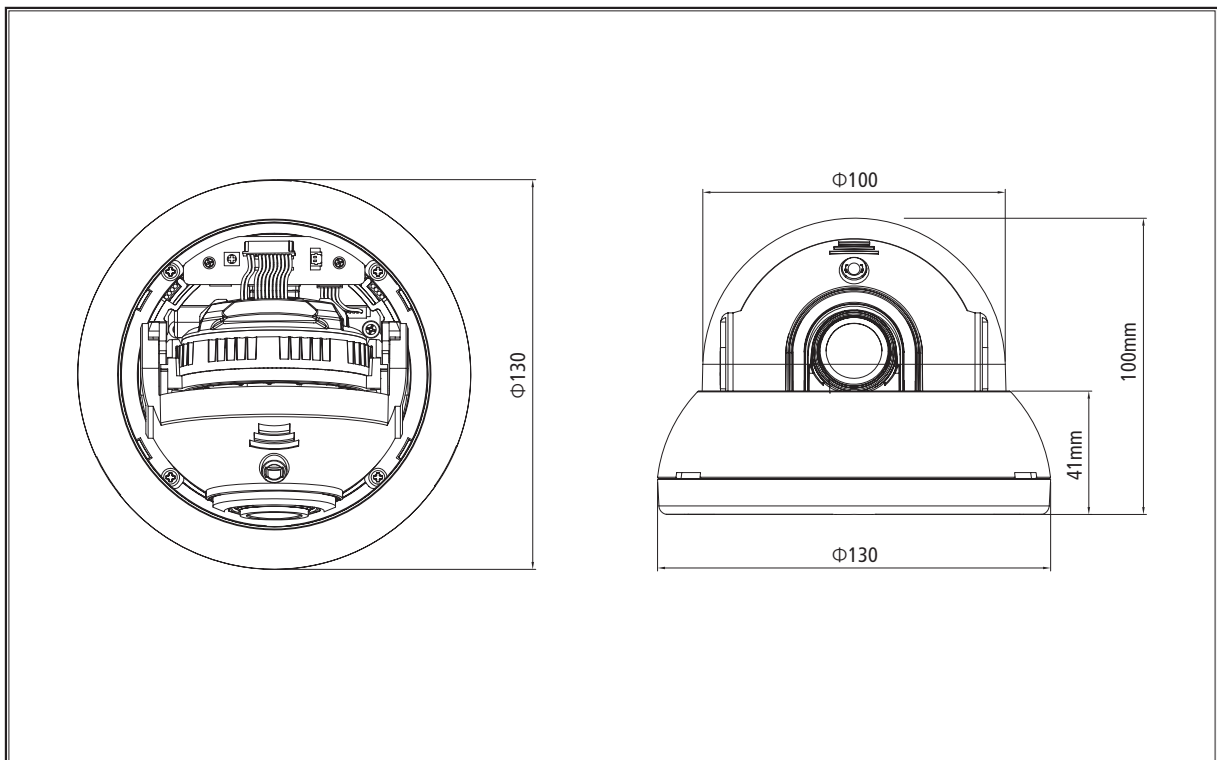


Camera Dimensions

5. W96MDVIR(W)

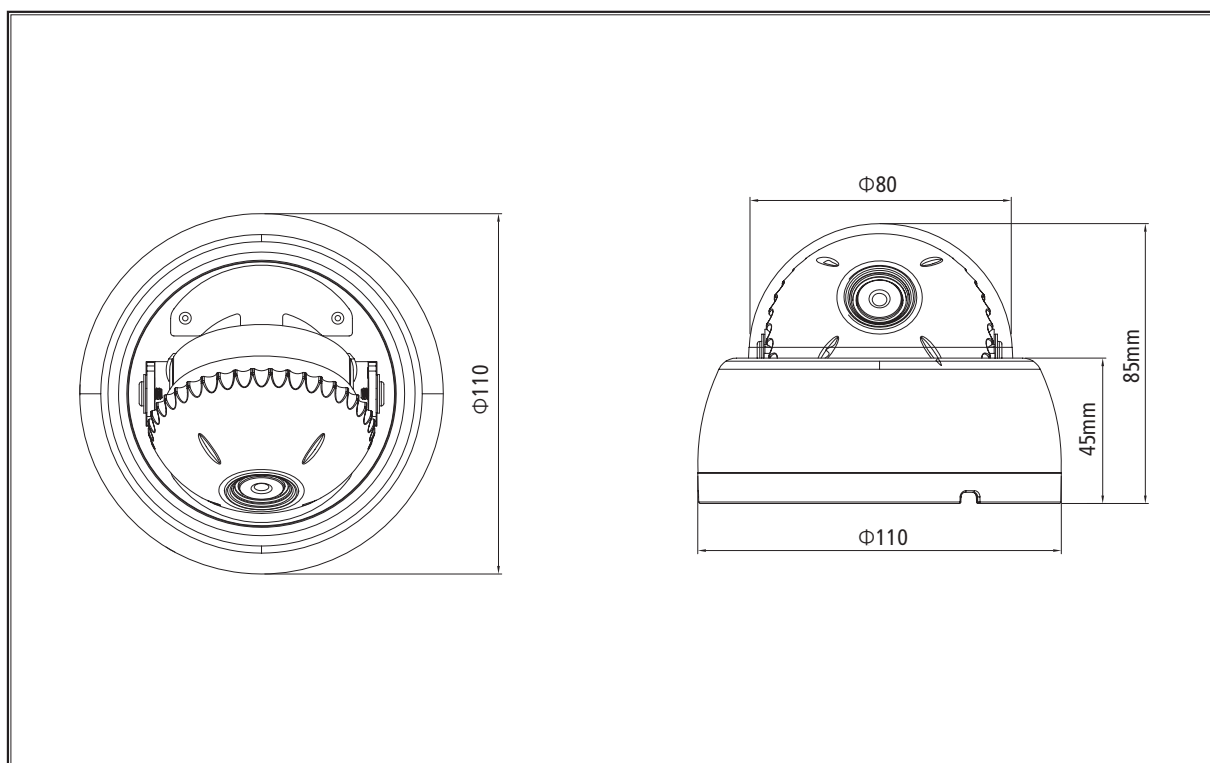


6. W96MDV(W)



Camera Dimensions

7. W96MDW



Troubleshooting

Problem	Possible Cause
Nothing appears on the screen.	Check the power cable, power supply output and video connection between the camera and monitor.
The image on the screen is dim.	- Are the camera lens or the lens glass dirty? (Clean the lens / glass with a soft clean cloth) - Adjust the monitor controls, as required. - If the camera is facing a very strong light, change the camera position. - Adjust the lens focus.
The image on the screen is dark.	- Adjust the contrast control of the monitor. - If there is an intermediate device, correctly set the 75 Ω/Hi-z.
The camera is not working properly and the surface of the camera is hot.	Check the camera is correctly connected to an appropriate regulated power source.
Motion Detection is not activated.	- Has MOTION DET been set to ON in the menu? - Has MD AREA been properly defined?
The colour of the picture is not correct.	Check the settings in WHITE BALANCE menu.
The image on the screen flickers.	Make sure that the camera isn't facing direct sunlight or fluorescent lighting. If necessary, change the camera position.
The SENS UP does not work.	- Check that the AGC setting in the EXPOSURE menu isn't set to OFF. - Check that the EXPOSURE menu and make sure SHUTTER is set to -----.



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